Question	Answer	Marks	Partial Marks
5(b)	6^{n-2} oe final answer	**2**	**B1** for answer of form 6^k oe or answer of the form $\left(\frac{1}{6}\right)^{-k}$ oe or for correct answer seen
5(c)	$3x^3 + 2x^2 - 37x + 12$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified or for simplified four-term expression of correct form with 3 terms correct or **B1** for correct expansion of two brackets with at least 3 terms out of 4 correct
5(d)(i)	eliminates the fraction correctly eg $(3x + 5)(x - 2) + 7 = x(x - 2)$	**M1**	
	$3x^2 + 5x - 6x - 10 + 7 = x^2 - 2x$ oe	**B2**	**B1** for $3x^2 + 5x - 6x - 10$ [+ 7] oe seen with at least 3 terms correct
	leading to $2x^2 + x - 3 = 0$	**A1**	**dep** on M1 B2 with no errors or omissions
5(d)(ii)	$(2x+3)(x-1)$	**M2**	or **M1** for $(2x + a)(x + b)$ where $ab = -3$ or $2b + a = [+]1$ or for partial factors $2x(x - 1) + 3(x - 1)$ or $x(2x + 3) - [1](2x + 3)$
	-1.5 oe and $+1$	**B1**	
5(e)	[TSA cylinder =] $2\pi x^2 + 2\pi x \times 3x$	**M1**	
	[TSA hemisphere=] $\pi(5y)^2 + \frac{4\pi(5y)^2}{2}$	**M1**	
	Leading to $2\pi x^2 + 6\pi x^2 = 50\pi y^2 + 25\pi y^2$ oe	**M1**	dep **M1M1**
	$x^2 = \frac{75y^2}{8}$	**A1**	dep on **M1M1M1**
6(a)	$\sqrt{10.4^2 + 6.5^2 - 2 \times 10.4 \times 6.5 \times \cos 64}$	**M2**	**M1** for $10.4^2 + 6.5^2 - 2 \times 10.4 \times 6.5 \times \cos 64$ **A1** for 91.1 to 91.2
	9.546 to 9.547	**A1**	

Question	Answer	Marks	Partial Marks
6(b)(i)	$180-(26+42)$	**B1**	
6(b)(ii)	6.89 or 6.888 to 6.892...	**3**	**M2** for $\frac{9.55}{\sin 112}\times\sin 42$ oe or **M1** for $\frac{\sin 112}{9.55}=\frac{\sin 42}{CD}$ oe
6(c)	5.84[2…]	**3**	**M2** for $\frac{x}{6.5}=\sin 64$ oe or **M1** for identifying shortest distance from D is perpendicular to AB
7(a)	2	**2**	**M1** for $3x+4x=6+8$ or better
7(b)	$5a(2a+1)$ final answer	**2**	**B1** for $a(10a+5)$ or $5(2a^2+a)$ or $5a(2a+1)$ then spoilt
7(c)	$4x(x-3)$ final answer	**2**	**M1** for $((2x-3)-3)((2x-3)+3)$ or better or for $4x^2-6x-6x+9$ [–9] oe or better
7(d)(i)	$\frac{1}{15}$ oe	**1**	
7(d)(ii)	19 683	**2**	**B1** for g(9), 3^9 or 3^{3^x} seen
7(d)(iii)	–3	**2**	**M1** for $3^k=\frac{1}{27}$ or $3^k=3^{-3}$ or answer g(–3)
8(a)	$y<10$ $y<x$ oe $x+y\leqslant 24$ oe	**3**	**B1** for each If 0 scored, **SC1** for $y\leqslant 10$ and $y\leqslant x$ and $x+y<24$